AP

FIR VisionAI

Complete System Architecture

& Step-by-Step User Guide

Andhra Pradesh Prohibition & Excise Department

FIR Sahayak u-FIR v2.0

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Classification: OFFICIAL — For Authorised Officers Only

Repository: github.com/doddasurendra/fir-sahayak

Generated from verified source code inspection

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- • Investigation Tools
- • End-to-End Six-Stage Workflow
- • Security, Roles & Audit
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- • Glossary

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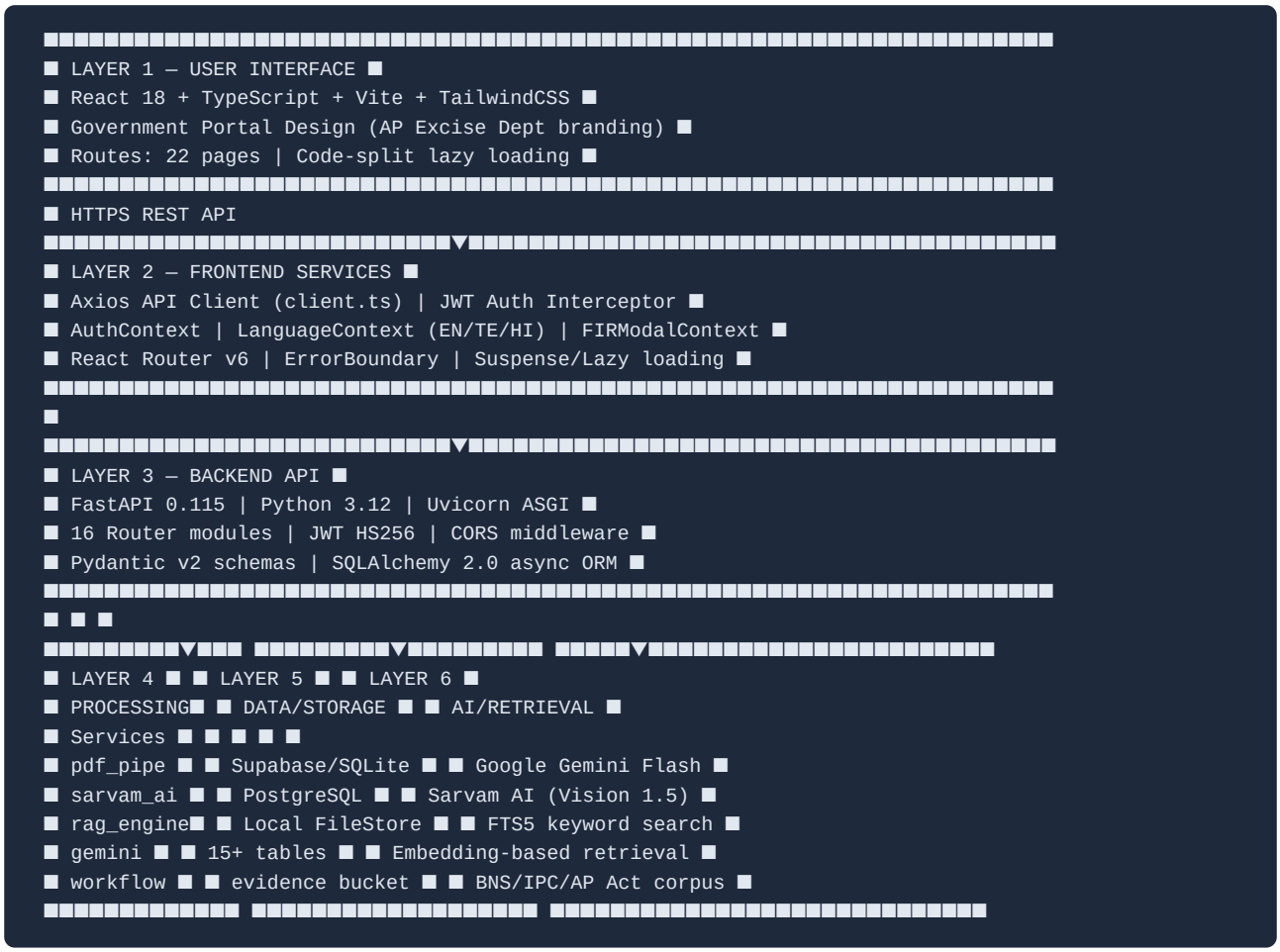
1. EXECUTIVE SUMMARY

FIR VisionAI (FIR Sahayak u-FIR v2.0) is a secure, AI-powered case intelligence platform developed for the **Andhra Pradesh Prohibition & Excise Department**. It digitises handwritten and printed First Information Reports (FIRs), performs multilingual OCR and structured field extraction, provides legal section intelligence, and enables officers to query case knowledge through an AI chatbot grounded exclusively in authorised documents.

Key Capabilities (Implemented & Verified)

Capability	Status
Multilingual FIR upload (PDF + image)	Implemented
AI text extraction via Gemini + Sarvam AI	Implemented
Officer verification & field override	Implemented
Legal section suggestion (BNS/IPC/AP Excise Act)	Implemented
AI FIR Chatbot — 4 grounded modes	Implemented
Knowledge Base management	Implemented
Historical FIR retrieval	Implemented
Entity network analysis	Implemented
Case diary & chargesheet generation	Implemented
Bulk document ingestion	Implemented
Cryptographic audit trail	Implemented
FIR folder classification (auto + manual)	Implemented
Dashboard KPI cards with navigation	Implemented
Role-based access (STATION / DISTRICT / STATE)	Implemented
Supabase PostgreSQL + local SQLite fallback	Implemented
PDF/XLSX/DOCX case export	Implemented

2. SYSTEM ARCHITECTURE OVERVIEW



Data Flow — FIR Upload to Answer

```
Officer uploads FIR PDF
↓
Frontend validates file (MIME, size)
↓
POST /api/v1/documents/upload → StorageService
↓
File stored: Supabase bucket OR local data/evidence/
↓
SHA-256 hash computed → Document record created in DB
↓
PDFPipeline.process_document() → pypdf extracts text per page
↓
DocumentPage records created with raw_text
↓
Text chunked (500 chars, 100 overlap) → DocumentChunk records
↓
FTS5 index updated (SQLite) OR pg_tsvector (PostgreSQL)
↓
GeminiService.extract_structured_case() → CaseField records
↓
SarvamDocAI.digitise_document() → OCR for scanned/handwritten
↓
Officer reviews in DocumentExtractionWorkspace
↓
Field overrides saved with verification_status = VERIFIED
↓
Case status: DRAFT → IN_REVIEW → APPROVED
↓
RAGEngine answers chatbot queries from DocumentChunk records
```

---

3. TECHNOLOGY STACK

Frontend

Technology	Version	Purpose
React	18.x	UI framework
TypeScript	5.x	Type safety
Vite	5.4	Build tool & dev server
TailwindCSS	3.x	Utility CSS styling
React Router	6.x	Client-side routing
Axios	1.x	HTTP client
Lucide React	Latest	Icon library
Recharts	2.x	Analytics charts

Backend

Technology	Version	Purpose
FastAPI	0.115	ASGI REST API framework
Python	3.12	Runtime
SQLAlchemy	2.0	Async ORM
Pydantic	v2	Schema validation
Uvicorn	Latest	ASGI server
pypdf	Latest	PDF text extraction
google-genai	Latest	Gemini AI SDK
httpx	Latest	Async HTTP (Sarvam AI)
argon2-cffi	Latest	Password hashing
PyJWT	Latest	JWT tokens
ReportLab	5.0.1	PDF generation
python-docx	Latest	DOCX generation
openpyxl	Latest	XLSX export

Database & Storage

Component	Details
Primary DB	Supabase PostgreSQL (cloud)
Fallback DB	SQLite 3 + FTS5 (local)
File Storage (Cloud)	Supabase Private Storage bucket "evidence"
File Storage (Local)	data/evidence/ directory
AI Provider	Google Gemini Flash (configurable)
OCR Provider	Sarvam AI Document Intelligence (Vision 1.5)

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4. FRONTEND ARCHITECTURE — ALL PAGES & FEATURES

Application Layout

Every page shares a consistent government portal layout:

- • **GovernmentHeader** — Official AP Government banner with Emblem, department name (AP Prohibition & Excise), Jai Hind badge, location pill (station/district), and language switcher (EN/TE/HI).
- • **HorizontalNav** — Dropdown navigation bar with all module access.
- • **Main Content Area** — Page-specific content (max-width 1700px, responsive).
- • **GovernmentFooter** — Statutory disclaimers, version info, copyright.

4.1 Login Page (`/login` — implicit, shown when unauthenticated)

Route: Rendered by `AppContent` when `isAuthenticated === false`

Component: `Login.tsx`

Features:

- • Username and password form fields
- • POST `/api/v1/auth/login` with credentials
- • JWT access token stored in `localStorage`
- • Refresh token support
- • Error display for invalid credentials
- • Auto-redirect to Dashboard on success

4.2 Dashboard (`/`)

Component: `Dashboard.tsx`

API: `GET /api/v1/analytics/dashboard`

KPI Cards (all clickable, URL-bound):

Card	Count	Target Route	Filter
Total Scoped Cases	41	`/cases`	All active cases
Under Review	28	`/cases?status=IN_REVIEW`	Awaiting verification
Approved FIRs	6	`/cases?status=APPROVED`	Court-ready records
Pending Drafts	7	`/cases?status=DRAFT`	Unsubmitted drafts

Charts:

- • Contraband Seizures by Volume (Bar chart — Recharts)
- • District-wise Case Distribution (Bar chart)

Quick Action Cards:

- • New Case Intake → `/intake`
- • AI FIR Assistance → `/ai-fir-assistance`
- • Legal Intelligence → `/legal`
- • Case Network → `/network`

Action Queue:

- • Pending Field Verifications (from `case_fields` table, `verification_status = UNVERIFIED`)
- • Recent Cases list with FIR numbers and statuses

4.3 Case Management (`/cases`)

Component: `CaseManagement.tsx`

API: `GET /api/v1/cases/?status={STATUS}&search={TERM}&limit=100`

Features:

- • URL-bound status filter (persists across browser Back/Forward/Refresh)
- • Quick-filter pills: All Cases | Under Review | Approved FIRs | Pending Drafts | Archived
- • Text search by FIR Number, Contraband Type, Offence Nature, Vehicle, Location
- • Table columns: FIR Number, Jurisdiction, Offence Nature, Contraband Seized, Vehicle Details, Status, Actions
- • Context-aware action buttons:
- • DRAFT → "Continue Draft" (Edit icon)
- • IN_REVIEW → "Review Case" (Eye icon)
- • APPROVED → "View Details" (Eye icon)
- • Date-Wise FIR Search component (by date range)
- • New Case Intake button (opens FIRFormSelectionModal)
- • Filter count banner showing active filter description
- • X-Total-Count header used for real record counts

4.4 FIR Folders (`/folders/:slug`)

Component: `FolderView.tsx`

API: `GET /api/v1/folders/{slug}/cases`

21 pre-seeded folders across 3 groups:

FIR Types: Illicit Distillation, NDPS/Drug Transport, Non-Duty-Paid Liquor, Vehicle Seizure, Unlawful Possession, Theft/Property Crime, Assault/Bodily Harm, Road/Traffic Accidents, Cheating/Fraud, Public Nuisance, Missing Persons, Sexual Offence (District+), Domestic Violence, Land/Property Dispute, Cyber Crime, Death Investigation, Arms & Explosives (District+)

Person-Based: Single-FIR Persons, Multiple-FIR Person Records

Status: Pending Classification, Under Investigation, Closed/Charge-sheeted

Features:

- • Auto-classification by keyword rules (classification.py service)
- • Manual folder assignment by officers
- • Folder-level search and status filter
- • AI confidence score display per case

4.5 Case Detail (`/cases/:id`)

Component: `CaseDetail.tsx`

API: `GET /api/v1/cases/{case_id}`

Sections:

- • Case header (FIR Number, District, Station, Registration Date, Status badge)
- • Extracted field table with confidence scores and verification status

- • Document list with Open Evidence Viewer button per document
- • Legal sections (AI-recommended + officer-accepted)
- • Entity list (Persons, Vehicles, Locations)
- • Workflow stage progress (CLASSIFICATION, ENTITY_EXTRACTION, AI_SUMMARY)
- • AI-generated case summary
- • Case Diary and Chargesheet generation buttons
- • Approve Case button (STATE/DISTRICT roles)
- • Archive Case button with reason field
- • Export buttons: XLSX, DOCX, CSV, JSON, PDF

Field Override Flow:

- • Officer clicks "Override" on a field
- • Modal opens with original vs. current value
- • Officer enters correction
- • PUT `/api/v1/cases/{case_id}/fields/{field_id}/override`
- • `verification_status` → OVERRIDDEN, `verified_by` set, audit log created

4.6 New Case Intake (`/intake`)

Component: `NewCaseIntake.tsx`

API: `POST /api/v1/documents/upload`

Upload Flow:

- • Officer opens page
- • Selects file (PDF or image: PNG, JPG, TIFF, WebP — max 50 MB)
- • Optional: link to existing Case ID
- • Click "Upload & Process"
- • Frontend POSTs multipart form to `/api/v1/documents/upload`
- • Backend validates MIME type, computes SHA-256, stores file
- • `Document` record created (status: PENDING → PROCESSING)
- • `PDFPipeline.process_document()` runs:
- • • PDF: text extracted per page via pypdf
- • • Image: sent to Sarvam AI for OCR + digitisation
- • `DocumentPage` records created with raw_text and OCR confidence
- • Text chunked and indexed in `document_chunks` + FTS5
- • Gemini extraction runs → `CaseField` records created
- • Status → COMPLETED
- • Officer redirected to Document Extraction Workspace

4.7 FIR Application Form (`/fir-application`, `/fir-application/:case_id`)

Component: `FIRApplicationForm.tsx`

API: `POST /api/v1/fir-applications/`, `GET /api/v1/fir-applications/{case_id}`

A structured digital FIR application form with:

- • Template selection (AP Excise, Standard FIR)
- • Language selection (English, Telugu, Hindi)
- • Fields: FIR Number, District, Station, Date, Accused Details, Offence Nature, Sections, Narrative

- • Draft save → Resume later
- • Submit for registration

4.8 Document Extraction Workspace (`/document-extraction`)

Component: `DocumentExtractionWorkspace.tsx`

APIs: `GET /api/v1/extraction/config`, `POST /api/v1/extraction/run`, `GET /api/v1/documents/{id}/pages`

Three-panel layout:

- • Left: Document list and extraction config
- • Center: Extracted field review table with confidence scores
- • Right: Document viewer (PDF or image)

Features:

- • Run extraction with current config
- • Field-by-field verification (VERIFY / OVERRIDE buttons)
- • Confidence score colour coding (Green ≥0.85 / Amber 0.6-0.84 / Red <0.6)
- • XLSX/DOCX export of extracted fields
- • Schema editor for custom extraction fields
- • Sarvam AI Extract job status polling

4.9 AI & Legal Intelligence (`/legal`)

Component: `LegalSectionAssistance.tsx`

API: `GET /api/v1/legal/suggest/{case_id}`, `POST /api/v1/legal/sections/{case_id}/accept`

Features:

- • AI-recommended legal sections for a case
- • Section details: Act name, Section number, Title, Full text, Penalty
- • Essential ingredients checklist
- • Matched facts from FIR text
- • Missing elements warning
- • Officer decision buttons: ACCEPT / REJECT (with reason)
- • Accepted sections carry to Chargesheet
- • Source: Verified corpus of AP Excise Act 1968, BNS 2023, IPC (historical), NDPS Act

> ***Important:*** *AI recommendations are suggestions only. An authorised officer must explicitly ACCEPT each section before it becomes part of the official case record.*

4.10 AI Case Summaries (`/summaries`)

Component: `AICaseSummaries.tsx`

API: `POST /api/v1/generation/summary/{case_id}`

- • Generates a factual AI summary of a case in EN/TE/HI
- • Grounded strictly in extracted and officer-verified case fields
- • Saved to `case_summaries` table with version tracking
- • Used as input for case diary generation

4.11 AI FIR Chatbot (`/ai-fir-assistance`)

Component: `AIFIRAssistance.tsx`

API: `POST /api/v1/rag/query`

Language Selection:

- • English | ■■■■■■■ | ■■■■■■■
- • Confirmation message sent in selected language

Source Modes:

Mode	Code	Data Source	FIR Required?
Selected FIR Only	PDF_ONLY	Single selected document's chunks	Yes
Knowledge Base Only	KNOWLEDGE_BASE_ONLY	KB documents (acts, circulars, guides)	No
FIR + Legal Corpus	FIR_PLUS_LEGAL	Selected FIR + legal_corpus table	Yes
Cross-Case Intelligence	CROSS_CASE	All authorized cases in jurisdiction	No

Document Selector:

- • Dropdown: Current case documents
- • Historical FIR documents (from `documents` table, `doc_category = HISTORICAL_FIR`)

Chat Features:

- • Message history with sender (AI/Officer) and timestamp
- • Citation cards showing physical page numbers and source document
- • "Open Evidence" button links citation to PDF Viewer Modal
- • Suggested questions per mode
- • Clear Chat button
- • Direct PDF upload for Mode A (bypasses case requirement)
- • Loading spinner during response generation

Response Format:

- • Answer text (grounded, never fabricated)
- • Citations: [{document_name, physical_page, excerpt}]
- • If not found: "Information not found in selected source" — never guesses

4.12 Knowledge Base (`/knowledge-base`)

Component: `KnowledgeBase.tsx`

APIs: `GET /api/v1/legal/kb`, `POST /api/v1/documents/upload` (with doc_category)

Document Categories:

- • LEGAL_ACT — Legislation texts
- • CIRCULAR — Department circulars

- • INVESTIGATION_GUIDE — Procedural manuals
- • REFERENCE — Reference materials
- • SEIZURE_REPORT — Historical seizure reports

Features:

- • Upload new KB document (PDF/image)
- • Auto-indexing into `document_chunks` after upload
- • Search by title, category, department
- • Document status: PENDING → PROCESSING → COMPLETED / FAILED
- • Re-index button for failed documents
- • Access control: DISTRICT+ role for restricted categories

4.13 Entity Network (`/network`)

Component: `EntityNetwork.tsx`

API: `GET /api/v1/entities/network/{case_id}`

- • Visual graph of entity relationships (NetworkGraph component using D3/force-directed)
- • Nodes: PERSON, VEHICLE, LOCATION, ORGANIZATION
- • Edges: CO_ACCUSED, VEHICLE_USED, ASSOCIATE
- • Community detection (Louvain algorithm — Planned for full implementation)
- • Risk score badges
- • Click node to view entity details and linked cases

4.14 Entity Review Queue (`/entity-review`)

Component: `EntityReviewQueue.tsx`

API: `GET /api/v1/entities/review-queue`, `PUT /api/v1/entities/{id}/merge`

- • Lists entities with `verification_status = REVIEW_PENDING`
- • Officers can CONFIRM, MERGE (deduplication), or REJECT entities
- • Merger links duplicate entity records into canonical entity

4.15 Case Diary Generator (`/case-diary`)

Component: `CaseDiaryGenerator.tsx`

API: `POST /api/v1/generation/case-diary/{case_id}`, `GET /api/v1/generation/documents`

- • Select case → Generate AI-drafted case diary
- • Content pulled from verified case fields, entities, and extracted narrative
- • Edit in rich-text editor before submitting for review
- • Status: DRAFT → IN_REVIEW → APPROVED
- • Export as DOCX or PDF once approved

4.16 Chargesheet Generator (`/chargesheet`)

Component: `ChargesheetGenerator.tsx`

API: `POST /api/v1/generation/chargesheet/{case_id}`

- • Select case with APPROVED legal sections
- • Generates structured chargesheet with:
- • FIR details
- • Accused particulars
- • Offence description
- • Invoked legal sections (officer-accepted only)
- • Evidence summary
- • Same review/approve workflow as case diary

4.17 Bulk Ingestion (`/bulk-ingestion`)

Component: `BulkIngestion.tsx`

API: `POST /api/v1/bulk/upload`, `GET /api/v1/bulk/status/{job_id}`

- • Upload CSV/JSON file with multiple FIR records
- • Background job processing with progress polling
- • Success/failure counts per record
- • Error report download

4.18 Employee Management (`/employees`)

Component: `EmployeeManagement.tsx`

APIs: `GET /api/v1/employees/`, `POST /api/v1/employees/`, `PUT /api/v1/employees/{id}`

- • User list with role, designation, station
- • Create new user account (STATE role required)
- • Edit user details and jurisdiction
- • Approval workflow: PENDING_APPROVAL → APPROVED
- • Activate/deactivate accounts

4.19 Audit Trail (`/audit-trail`)

Component: `AuditTrail.tsx`

API: `GET /api/v1/audit/logs`

- • Cryptographically hash-chained audit log (SHA-256)
- • Filters: Actor, Action Type, Object Type, Date Range
- • Each entry shows: Timestamp, Actor, Action, Object, Before/After state
- • Tamper-evident: each entry's hash includes previous entry hash (blockchain-style)

4.20 Reports & Analytics (`/reports`)

Component: `ReportsAnalytics.tsx`

API: `GET /api/v1/analytics/dashboard`

- • Contraband seizure statistics
- • District-wise case distribution
- • Status pipeline chart
- • Export report as XLSX

4.21 Settings & Health (`/settings`)

Component: `SettingsHealth.tsx`

APIs: `GET /api/v1/health`, `GET /api/v1/system/backups`, `POST /api/v1/system/backup`

- • System health dashboard (FastAPI status, Database, AI Model)
- • Create database backup snapshot (SQLite integrity verification + SHA-256)
- • Download existing backup files
- • Documentation download buttons (Architecture Guide PDF, Technical Report, Presentation)

4.22 Persons View (`/folders/persons/:type`)

Component: `PersonsView.tsx`

APIs: `GET /api/v1/folders/persons/single-fir`, `GET /api/v1/folders/persons/multiple-fir`

- • Single-FIR Persons: Entities linked to exactly 1 case
- • Multiple-FIR Persons: Potential repeat offenders (2+ cases)
- • Search by name, risk score filter
- • Links to entity detail and associated cases

> ***IMPORTANT:*** *Multiple-case matches are advisory indicators only. Similar names or details across cases must not be treated as proof that two records involve the same individual without officer verification.*

---

5. BACKEND ARCHITECTURE — ALL API ENDPOINTS

FastAPI application entry point: `backend/app/main.py`

Base URL: `http://localhost:8000/api/v1`

5.1 Authentication (`/auth`)

Method	Endpoint	Purpose
POST	`/auth/login`	Username/password login → access + refresh tokens
POST	`/auth/refresh`	Refresh access token using refresh token
POST	`/auth/logout`	Revoke refresh token
GET	`/auth/me`	Current user profile
POST	`/auth/register`	Register new user (pending approval)

Auth mechanism: JWT HS256 tokens. Access token expires in 120 minutes. Refresh token expires in 7 days. Both stored on frontend. `get_current_user()` dependency validates every protected route.

5.2 Cases (`/cases`)

Method	Endpoint	Purpose
GET	`/cases/`	List cases (filters: status, district, station, search)
GET	`/cases/{case_id}`	Full case detail with fields, entities, documents
POST	`/cases/`	Create new case record
PUT	`/cases/{case_id}`	Update case fields
POST	`/cases/{case_id}/approve`	Approve case (DISTRICT+ role)
POST	`/cases/{case_id}/archive`	Archive/soft-delete case
GET	`/cases/{case_id}/export/{format}`	Export: xlsx/docx/csv/json/pdf
PUT	`/cases/{case_id}/fields/{field_id}/override`	Officer field override

5.3 Documents (`/documents`)

Method	Endpoint	Purpose
POST	`/documents/upload`	Upload PDF or image evidence
GET	`/documents/{doc_id}`	Document metadata
GET	`/documents/{doc_id}/pages`	All pages with OCR text
GET	`/documents/{doc_id}/view`	Serve file (signed URL or local stream)
GET	`/documents/{doc_id}/view-page/{page}`	Serve single page image
POST	`/documents/{doc_id}/replace`	Replace document with new version
POST	`/documents/{doc_id}/archive`	Archive document
GET	`/documents/{doc_id}/signed-url`	Get time-limited download URL

5.4 Extraction (`/extraction`)

Method	Endpoint	Purpose
POST	`/extraction/run`	Run AI extraction on a document

Method	Endpoint	Purpose
GET	`/extraction/runs/{doc_id}`	List extraction runs for a document
GET	`/extraction/config`	Get active extraction configuration
PUT	`/extraction/config`	Update extraction schema/instructions
POST	`/extraction/narrative`	Extract from free-text narrative
GET	`/extraction/export/{run_id}/xlsx`	Export extraction as XLSX
GET	`/extraction/export/{run_id}/docx`	Export extraction as DOCX

5.5 Legal Intelligence (`/legal`)

Method	Endpoint	Purpose
GET	`/legal/corpus`	List all legal corpus sections
GET	`/legal/corpus/{section_id}`	Single section detail
GET	`/legal/suggest/{case_id}`	AI-suggested sections for a case
POST	`/legal/sections/{case_id}/accept`	Officer accepts a section
POST	`/legal/sections/{case_id}/reject`	Officer rejects a section
GET	`/legal/kb`	Knowledge Base document list
POST	`/legal/kb/upload`	Upload KB document
POST	`/legal/kb/{doc_id}/reindex`	Re-index failed KB document

5.6 RAG / AI Chatbot (`/rag`)

Method	Endpoint	Purpose
POST	`/rag/query`	Submit question to RAG engine

Request body:

```
{
"query": "What contraband was seized?",
"mode": "PDF_ONLY | KNOWLEDGE_BASE_ONLY | FIR_PLUS_LEGAL | CROSS_CASE",
"document_id": "uuid-of-selected-document",
"language": "en | te | hi"
}
```

Response:

```
{
"answer": "Grounded answer text...",
"citations": [
{
"document_name": "FIR_2024_KNL_001.pdf",
"physical_page": 2,
"excerpt": "...relevant text snippet..."
}
],
"mode": "PDF_ONLY",
"confidence": 0.87
}
```

5.7 Entities (`/entities`)

Method	Endpoint	Purpose
GET	`/entities/`	List entities (filter by type, case)
GET	`/entities/{entity_id}`	Entity detail with all mentions
GET	`/entities/network/{case_id}`	Graph edges for case
GET	`/entities/review-queue`	Entities pending verification
PUT	`/entities/{entity_id}/merge`	Merge duplicate entities
PUT	`/entities/{entity_id}/verify`	Confirm entity
GET	`/entities/repeat-offenders`	Entities with 2+ case mentions

5.8 Generation (`/generation`)

Method	Endpoint	Purpose
POST	`/generation/summary/{case_id}`	Generate AI case summary
POST	`/generation/case-diary/{case_id}`	Generate case diary draft
POST	`/generation/chargesheet/{case_id}`	Generate chargesheet draft

Method	Endpoint	Purpose
GET	`/generation/documents`	List all generated documents
PUT	`/generation/documents/{doc_id}`	Edit generated document
POST	`/generation/documents/{doc_id}/approve`	Approve generated document
GET	`/generation/documents/{doc_id}/export/pdf`	Export as PDF

5.9 Bulk Ingestion (`/bulk`)

Method	Endpoint	Purpose
POST	`/bulk/upload`	Upload CSV/JSON for batch processing
GET	`/bulk/status/{job_id}`	Poll job status
GET	`/bulk/jobs`	List all bulk jobs

5.10 Employees (`/employees`)

Method	Endpoint	Purpose
GET	`/employees/`	List employees
POST	`/employees/`	Create employee account
PUT	`/employees/{id}`	Update employee
PUT	`/employees/{id}/approve`	Approve pending account
PUT	`/employees/{id}/deactivate`	Deactivate account

5.11 Audit (`/audit`)

Method	Endpoint	Purpose
GET	`/audit/logs`	Paginated audit log with filters
GET	`/audit/logs/{entry_uuid}`	Single audit entry

5.12 Analytics (`/analytics`)

Method	Endpoint	Purpose
GET	`/analytics/dashboard`	KPI counts, contraband stats, district stats
GET	`/analytics/repeat-offenders`	Repeat offender analysis

5.13 Folders (`/folders`)

Method	Endpoint	Purpose
GET	`/folders/`	List all FIR folder categories
GET	`/folders/counts`	Case count per folder (scoped)
GET	`/folders/{slug}/cases`	Cases in a specific folder
POST	`/folders/{slug}/cases/{case_id}`	Manually assign case to folder
GET	`/folders/workflow/{case_id}`	Workflow execution stages
GET	`/folders/persons/single-fir`	Persons with exactly 1 case
GET	`/folders/persons/multiple-fir`	Persons with 2+ cases

5.14 FIR Applications (`/fir-applications`)

Method	Endpoint	Purpose
POST	`/fir-applications/`	Submit new FIR application form
GET	`/fir-applications/{case_id}`	Get application for case
PUT	`/fir-applications/{case_id}`	Update draft application

5.15 System (`/system`)

Method	Endpoint	Purpose
POST	`/system/backup`	Create database backup snapshot
GET	`/system/backups`	List backup files
GET	`/system/backups/download/{filename}`	Download backup file

5.16 Download Endpoints (root level)

Method	Endpoint	Purpose
GET	`/api/v1/health`	System health check
GET	`/download/report`	Download Technical Report (.docx)
GET	`/download/presentation`	Download Presentation (.pptx)
GET	`/download/guide`	Download Architecture & User Guide (.pdf)

5.17 Jurisdictions (`/jurisdictions`)

Method	Endpoint	Purpose
GET	`/jurisdictions/districts`	List all AP districts
GET	`/jurisdictions/stations/{district}`	List stations in district

---

6. DATABASE SCHEMA & RELATIONSHIPS

6.1 All Tables

Table	Purpose
`users`	Officer accounts with role and jurisdiction
`refresh_tokens`	JWT refresh token revocation
`cases`	FIR case records — core table
`case_fields`	Extracted field values per case with confidence
`case_legal_sections`	AI-recommended + officer-accepted legal sections
`documents`	Uploaded evidence documents (PDF/images)
`document_pages`	Per-page OCR text and confidence
`document_chunks`	Text chunks for RAG retrieval (500 char, 100 overlap)
`entities`	Persons, Vehicles, Locations, Organizations
`entity_mentions`	Entity appearances in specific cases
`graph_edges`	Entity-entity relationships (CO_ACCUSED, VEHICLE_USED)
`legal_corpus`	Verified BNS/IPC/AP Excise Act sections

Table	Purpose
`case_summaries`	AI-generated case summaries (EN/TE/HI)
`generated_documents`	Case diary and chargesheet drafts
`document_approval_history`	Version history of generated docs
`audit_log`	Cryptographic hash-chained action log
`bulk_jobs`	Batch import job tracking
`fir_folders`	21 pre-seeded FIR category folders
`case_folder_assignments`	Many-to-many: cases ↔ folders
`workflow_executions`	Automated post-intake workflow stage tracking
`extraction_configs`	AI extraction schema configuration
`extraction_runs`	Per-document extraction job records
`stations` (optional)	District/station master data

6.2 Key Relationships

```
users ■■■
```

6.4 Document Lifecycle

```
PENDING ■■→ PROCESSING ■■→ COMPLETED
↓
FAILED ■■→ (retry)
```

6.5 Role-Based Data Scoping (apply_case_scope_filter)

Role	Visible Cases
STATION	Cases where district = user.district AND station = user.station
DISTRICT	Cases where district = user.district
STATE	All cases in the system

---

7. OCR & EXTRACTION PIPELINE

Stage 1 — File Intake & Validation

- • Frontend validates MIME type (PDF, PNG, JPG, TIFF, WebP) and file size (max 50 MB)
- • POST to `/api/v1/documents/upload` with multipart form
- • Backend: `_detect_mime_type()` verifies actual file header bytes
- • SHA-256 hash computed for deduplication
- • File stored: Supabase private storage bucket OR local `data/evidence/` directory

Stage 2 — PDF Text Extraction (Digital PDFs)

- • `PDFPipeline.process_pdf()` uses `pypdf.PdfReader`
- • Text extracted per page: `page.extract_text()`
- • `DocumentPage` records created with `raw_text`, `physical_page_number`
- • Page count stored in `Document.physical_page_count`

Stage 3 — OCR for Scanned/Handwritten Documents

Provider: Sarvam AI Document Intelligence (Vision 1.5)

- • `SarvamDocAI.digitise_document()`: submits document for full OCR
- • Returns: reading-order blocks, markdown content per page
- • Max 10 pages per job (splits larger documents automatically)
- • `SarvamDocAI.extract_document()`: schema-based structured extraction
- • Returns: key-value pairs with confidence scores and source citations
- • Jobs polled until COMPLETED/FAILED (terminal states)

- • Results stored in `DocumentPage.raw_text` and `ExtractionRun.raw_extracted_json`

Fallback provider: Google Gemini Vision

- • Used when Sarvam AI is unconfigured or fails
- • Gemini vision model reads image/PDF as multimodal input

Stage 4 — Structured Field Extraction

Service: `GeminiService.extract_structured_case()`

- • Input: Raw text from OCR/PDF extraction
- • Prompt: Strict defensive extraction rules (no assumption of offence type)
- • Output: JSON with 16 standard FIR fields:
- • • fir_number, year, district, station, mandal, village
- • • offence_date, detection_date, registration_date
- • • offence_nature, contraband_type, seized_quantity, seized_unit
- • • accused_names[], witness_names[], seizing_officer_name
- • • vehicle_details, law_sections[], narrative_summary
- • `CaseField` records created with `confidence_score` and `extraction_method`
- • Null returned for any field not explicitly present in source text

Stage 5 — Text Chunking & Indexing

- • `chunk_text()`: 500-character chunks with 100-character overlap
- • `DocumentChunk` records created with `chunk_text`, `char_start`, `char_end`, `physical_page_number`
- • SQLite: FTS5 full-text search index populated
- • PostgreSQL: Indexed via `pg_trgm` and LIKE patterns
- • Embeddings: `DocumentChunk.embedding_json` column exists for future vector search

Stage 6 — Human-in-the-Loop Review

- • Fields with confidence < 0.85 flagged for officer review (reference threshold from HLD)
- • Officer views extracted fields in DocumentExtractionWorkspace
- • Each field shows: original_value, normalized_value, confidence_score, source page
- • VERIFY: marks `verification_status = VERIFIED`
- • OVERRIDE: officer enters correct value → `verification_status = OVERRIDDEN`, audit log entry

Stage 7 — Post-Intake Automation

- • `trigger_post_intake_workflow()` runs after case save
- • Stage 1: CLASSIFICATION — `classify_case_text()` assigns FIR folder via keyword rules
- • Stage 2: ENTITY_EXTRACTION — extracts Person/Vehicle entities (future expansion)
- • Stage 3: AI_SUMMARY — generates case summary (triggered on demand)
- • Each stage tracked in `workflow_executions` table with PENDING/RUNNING/COMPLETED/FAILED status

---

8. AI FIR CHATBOT & RAG PIPELINE

End-to-End Question Flow

```
1. Officer types question → selects mode + document (if needed)
2. POST /api/v1/rag/query {query, mode, document_id, language}
3. Backend: get_current_user() → validates JWT
4. Mode dispatched to RAGEngine method
5. Document authorization checked (case_id ownership via apply_case_scope_filter)
6. DocumentChunk records retrieved for selected document
7. Hybrid retrieval:
- SQLite: FTS5 MATCH query
- PostgreSQL: ILIKE pattern matching across chunks
- Full document fallback if sparse results
8. Retrieved chunks passed to GeminiService with grounding prompt
9. Gemini generates answer grounded ONLY in provided chunks
10. Response: {answer, citations[{document_name, physical_page, excerpt}]}
11. RAG query logged in audit_log (action_type = RAG_QUERY_EXECUTED)
12. Response returned to frontend
13. Citations rendered as clickable cards linking to PDF viewer
```

Mode A — Selected FIR Only (PDF_ONLY)

- • `RAGEngine.answer_from_active_fir(document_id, query, user, db, language)`
- • Strictly isolated to ONE document's chunks
- • Citation: physical page number within that document
- • In-memory cache: 5-minute TTL keyed by (document_id, query_hash, language)
- • Auto-reindexes chunks if missing

Mode B — Knowledge Base Only (KNOWLEDGE_BASE_ONLY)

- • `RAGEngine.answer_from_knowledge_base(query, user, db, language)`
- • Searches `document_chunks` where parent document has `doc_category IN (LEGAL_ACT, CIRCULAR, INVESTIGATION_GUIDE, REFERENCE)`
- • Jurisdiction-scoped: only documents accessible to user's role
- • No case documents included

Mode C — FIR + Legal Corpus (FIR_PLUS_LEGAL)

- • `RAGEngine.answer_fir_plus_legal(document_id, query, user, db, language)`
- • Combines Mode A document chunks with `legal_corpus` table sections
- • Both sources cited separately in response
- • Officers can ask: "Which BNS sections apply to the facts in this FIR?"

Mode D — Cross-Case Intelligence (CROSS_CASE)

- • `RAGEngine.answer_cross_case(query, user, db, language)`
- • Searches across ALL case documents within user's authorized jurisdiction
- • Useful for: "Have we seen this vehicle registration before?"
- • Role-scoped: STATION sees only their station's cases

> ***Grounding guarantee:*** *The Gemini prompt instructs the model: "Answer ONLY from the provided context. If the information is not present, say it was not found. Never invent facts, page numbers, FIR numbers, names, or legal sections."*

9. KNOWLEDGE BASE & LEGAL CORPUS

Legal Corpus (legal_corpus table)

Pre-seeded from JSON files at startup:

- • `ap_excise_act_1968.json` — AP Prohibition & Excise Act sections
- • `bns_2023.json` — Bharatiya Nyaya Sanhita 2023 sections
- • `ipc_historical.json` — IPC sections (historical reference)

Each entry contains:

- • act_name, section_number, section_title, full_text
- • essential_ingredients_json (checklist for prosecution)
- • penalty_text, effective_date, repeal_date
- • source_url, source_gazette (authoritative provenance)
- • verification_status = VERIFIED

Knowledge Base Documents

- • Uploaded via `/knowledge-base` page or `/api/v1/legal/kb/upload`
- • Document categories: LEGAL_ACT, CIRCULAR, INVESTIGATION_GUIDE, REFERENCE
- • After upload: same pipeline as evidence documents (OCR → chunks → indexed)
- • Status must be COMPLETED before chatbot can retrieve content
- • Re-index option for FAILED documents

10. INVESTIGATION TOOLS

Entity Resolution

Implemented:

- • Entities extracted during case processing (PERSON, VEHICLE, LOCATION, ORGANIZATION)
- • `entity_resolution.py` service: deduplication by name similarity
- • Aliases stored in `aliases_json` for variant spellings (e.g., "Ramaiah" / "Ramaiya")
- • Merge workflow: duplicate → REVIEW_PENDING → officer merges → merged_into_id set
- • Risk score computed based on number of case mentions

Planned/Partial:

- • Transliteration-based matching across Telugu/Hindi/English names
- • Automated phonetic deduplication (Soundex/Metaphone)

Network Graph

Implemented:

- • Force-directed graph visualization (NetworkGraph component)

- • Node types: PERSON (circle), VEHICLE (square), LOCATION (diamond)
- • Edge types: CO_ACCUSED, VEHICLE_USED, ASSOCIATE
- • Community IDs (Louvain algorithm output stored in `community_id` column)

Planned:

- • Live community detection calculation
- • Centrality scoring visualization

Repeat Offender Detection

Implemented:

- • Persons View (`/folders/persons/multiple-fir`) lists entities appearing in 2+ cases
- • `GET /api/v1/analytics/repeat-offenders` returns analysis
- • Risk factors listed (number of cases, roles — ACCUSED, ASSOCIATE)

CRITICAL WARNING: Similar names or vehicle numbers appearing in multiple cases are advisory indicators only. They must never be treated as proof of identity or guilt without full officer investigation and verification.

---

11. END-TO-END SIX-STAGE WORKFLOW

Example Case: Illicit Distillation FIR — Kurnool District

Stage 1 — Evidence Upload & Extraction

- • Officer opens New Case Intake (`/intake`)
- • Uploads scanned FIR PDF: `FIR_2024_KNL_001.pdf`
- • System stores file → creates Document record (PENDING)
- • PDFPipeline runs → text extracted per page
- • Sarvam AI OCR digitises handwritten sections
- • Gemini extracts: FIR Number, Accused Names, Contraband, Quantity, Sections
- • Document status → COMPLETED
- • Officer opens DocumentExtractionWorkspace
- • Reviews extracted fields, verifies accurate ones, overrides errors
- • Case created: `status = DRAFT`

Stage 2 — Officer Review & Sign-Off

- • Officer navigates to Case Management → case appears in DRAFT queue
- • Opens Case Detail
- • Reviews all extracted fields and confidence scores
- • Verifies entity records (accused names, vehicle numbers)
- • Reviews AI-suggested legal sections (AP Excise Act §34(a), BNS §304)
- • ACCEPTS applicable sections, REJECTS inapplicable ones
- • Submits for supervisor review: `status → IN_REVIEW`

Stage 3 — AI & Legal Intelligence

- • Supervisor opens case in Legal Intelligence (`/legal`)
- • Reviews AI-recommended sections with matched facts

- • Checks essential ingredients checklist
- • Accepts final legal sections
- • Generates AI Case Summary (EN/TE/HI as needed)
- • Reviews chatbot response to "What quantity of illicit liquor was seized and where?" (Mode A)

Stage 4 — Investigation Tools

- • Opens Entity Network (`/network`)
- • Sees accused person linked to 1 prior case (VEHICLE_USED edge)
- • Officer notes advisory match — initiates manual verification
- • Confirms entity identity through official records
- • Updates entity verification_status to CONFIRMED

Stage 5 — Administration & Existing FIRs

- • Supervisor approves case: `status` → `APPROVED`
- • Case diary generated and approved
- • Chargesheet generated with all accepted legal sections
- • Chargesheet exported as DOCX for court filing
- • All actions recorded in cryptographic audit_log

Stage 6 — Repeat Offender Intelligence *(advisory only)*

- • Monthly review: open Persons View → Multiple-FIR
- • Accused "Raju S/o Lakshmaiah" appears in 3 cases across 2 years
- • Risk score: 0.74 (based on frequency and recency)
- • Officers review all 3 cases for investigative patterns
- • Repeat-offender status: ADVISORY — requires officer verification before any official action

---

12. SECURITY, ROLES & AUDIT

Authentication

- • JWT HS256 tokens (access: 120 min, refresh: 7 days)
- • Argon2id password hashing (OWASP recommended)
- • Refresh token revocation on logout stored in `refresh_tokens` table
- • `is_revoked` flag prevents reuse of revoked tokens
- • All routes protected by `Depends(get_current_user)`

Role-Based Access Control

Role	Scope	Capabilities
STATION	Own station cases only	Create, upload, extract, verify, review
DISTRICT	All cases in district	All STATION actions + Approve, District reports

Role	Scope	Capabilities
STATE	All cases system-wide	All actions + Employee management, system config

Jurisdiction Enforcement

- • `apply_case_scope_filter()` applied on every case list query
- • STATION: `WHERE district = ? AND station = ?`
- • DISTRICT: `WHERE district = ?`
- • STATE: No filter (sees all)
- • Cannot be bypassed by URL manipulation — enforced at DB query level

Audit Logging

- • Every significant action logged in `audit_log`
- • SHA-256 hash chain: `entry_hash = SHA256(previous_hash + canonical_content)`
- • Genesis hash: 64 zeros
- • Tampering with any entry breaks the subsequent hash chain
- • Logged actions include: CASE_CREATED, FIELD_OVERRIDE, SECTION_ACCEPTED, SECTION_REJECTED, RAG_QUERY_EXECUTED, CASE_APPROVED, DOCUMENT_ARCHIVED, ENTITY_MERGED

Secure File Access

- • Supabase: Private bucket (not publicly accessible)
- • Signed URLs generated server-side with 1-hour expiry
- • Local fallback: Files served through authenticated FastAPI endpoint only
- • SHA-256 hash stored per document for integrity verification

---

13. TROUBLESHOOTING GUIDE

Error	Possible Cause	Resolution
Login fails (401)	Wrong credentials, account inactive	Verify username/password; check approval_status in users table
Upload fails (413)	File too large (>50 MB)	Compress or split document
Upload fails (415)	Unsupported file type	Use PDF, PNG, JPG, TIFF, or WebP
OCR fails	Sarvam AI key missing or expired	Check SARVAM_API_KEY in .env; system falls back to Gemini
Extraction returns nulls	Gemini API key missing or quota exceeded	Check GEMINI_API_KEY; system uses rule-based fallback

Error	Possible Cause	Resolution
Document stuck in PROCESSING	OCR job timeout	Re-upload document; check Sarvam API status
Historical FIR not found	document_id not selected in chatbot	Select document from dropdown in Mode A
Chatbot no answer	Chunks not indexed	Check document status = COMPLETED; use re-index option
Evidence image broken	Signed URL expired	Refresh page; backend generates new signed URL
Permission denied (403)	Role insufficient for action	Check user role; contact district/state admin
Database error	Supabase connection issue	Check SUPABASE_DATABASE_URL; system falls back to SQLite
PDF download fails (404)	PDF file not generated yet	Generate document first; check status = APPROVED
Case count mismatch	KPI cache stale (60s TTL)	Wait 60 seconds or restart backend
Incorrect citation	Chunks from different document in results	Verify document_id in chatbot selector

---

14. TESTING & DEPLOYMENT

Running Locally

Backend:

```
cd backend
python -m uvicorn app.main:app --host 0.0.0.0 --port 8000 --reload
```

Frontend:

```
cd frontend
npm install
npm run dev
# Opens at http://localhost:5173
```

Demo Accounts:

Username	Password	Role
developer_test	DevPass@2024	STATE
state_director	StateAdmin@2024	STATE
district_admin	DistrictPass@2024	DISTRICT — Kurnool
station_officer	StationPass@2024	STATION — Kurnool Town

Environment Variables (.env)

```
SUPABASE_URL=https://your-project.supabase.co
SUPABASE_KEY=your-anon-key
SUPABASE_SERVICE_ROLE_KEY=your-service-role-key
SUPABASE_DATABASE_URL=postgresql://postgres:password@db.project.supabase.co:5432/postgres
SUPABASE_STORAGE_BUCKET=evidence
GEMINI_API_KEY=your-gemini-api-key
SARVAM_API_KEY=your-sarvam-api-key
```

Build for Production

```
cd frontend
npm run build # Generates dist/ directory
```

```
cd backend
pip install -r requirements.txt
python -m uvicorn app.main:app --host 0.0.0.0 --port 8000 --workers 4
```

---

15. GLOSSARY

Term	Definition
BNS	Bharatiya Nyaya Sanhita 2023 — new criminal code replacing IPC
IPC	Indian Penal Code (historical — superseded by BNS)
FIR	First Information Report — initial police/excise complaint
NDPS	Narcotic Drugs and Psychotropic Substances Act
OCR	Optical Character Recognition

Term	Definition
RAG	Retrieval-Augmented Generation — AI answers grounded in source documents
FTS5	SQLite Full-Text Search version 5
RLS	Row Level Security (Supabase/PostgreSQL)
IO	Investigating Officer
HITL	Human-in-the-Loop — officer verification step
Chunk	500-character segment of extracted text used for RAG retrieval
Canonical Entity	The primary record after deduplication of entity aliases
Signed URL	Time-limited authenticated URL for accessing private stored files
JWT	JSON Web Token — authentication credential
SHA-256	Cryptographic hash algorithm used for file integrity and audit chaining
AP Excise	Andhra Pradesh Prohibition & Excise Department
Sarvam AI	Indian AI company providing document OCR/extraction API
Gemini	Google's multimodal AI model used for extraction and generation

---

Document generated from verified source code inspection of FIR Sahayak u-FIR v2.0.

Reference HLD: Multilingual Handwritten FIR Extraction Engine (Google Docs).

Features labelled "Planned" are not yet fully implemented in the codebase.